

ADRIAN TABOADA PEREZ

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EDUCATION

Carnegie Mellon University

M.S. in Mechanical Engineering, Concentration in Robotics and Control Systems

Pittsburgh, PA

Expected Dec 2027

Boston University

B.S. in Mechanical Engineering, Minor: Electrical Engineering (Cum Laude)

Boston, MA

Sep 2022 – May 2026

Relevant Courses: Energy Conversion, Power Electronics, Analog Circuits, Manufacturing Processes, Instrumentation

RELEVANT EXPERIENCE

Field Engineer Intern | Tesla

Austin, TX | May 2026 - Present

- Coordinated with subcontractors, suppliers and cross-functional teams to track daily equipment installation activities, ensuring on-time execution and clear communication across stakeholders.
- Created CAD release-ready manufacturing drawings using SolidWorks for custom parts and fixtures machines for production tooling, supporting tool installation timelines and build readiness.
- Performed hands-on modification and installation of manufacturing equipment components, ensuring compliance with company safety standards and technical specifications.
- Participated in site visits to monitor installation progress, verify safety compliance, and support on-site troubleshooting and documentation.

Electronics Lab Assistant and Teaching Assistant | Boston University

Boston, MA | Dec 2025 - May 2026

- Maintained and troubleshoot lab equipment (oscilloscopes, function generators, power supplies) and organized electronics components to support electronics courses, ensuring reliable operation during sessions.
- Served as Teaching Assistant for EC410 Electronics, guiding ~30 students per session through hands-on labs on transistor circuits (BJTs, MOSFETs), amplifiers, and filters, and grading lab reports and assignments.

Operations and Logistics Intern | Limpieza Alquimia S.A. de C.V.

Mexico City, Mexico | Jun 2025 - Aug 2025

- Analyzed delivery and operations data for an industrial cleaning products manufacturer to identify routing delays and workflow bottlenecks; delivered recommendations to improve weekly planning and delivery efficiency.
- Built Excel tools to track inventory levels and delivery schedules, enabling earlier low-stock detection and faster response to recurring supplier issues (PivotTables, XLOOKUP, SUMIFS, and conditional-formatting alerts).
- Calibrated industrial flow meters for multiple detergent solutions; created calibration profiles and trained plant operators on setup and usage to improve batch consistency.

SELECTED PROJECTS

Tube Swaging Automation Project | Vita Needle Company

Sep 2025 – May 2026

- Designed and built an automation device for a tube swaging manufacturing process using a lead-screw actuator, pneumatic gripper, and custom gripper jaws to grab and load needs into the swaging machine, replacing manual feeding process and achieving 98% consistent, high-quality swages.
- Developed the system's electrical control system and HMI, integrating a microcontroller, Hall effect sensors, and a solenoid-driven gripper, with a fail-safe E-STOP circuit using relays and contactors for reliable operator safety.

Centrifugal Fuel Pump | BU Rocket Propulsion Group

Oct 2025 - Dec 2025

- Modeled a centrifugal fuel pump, 3D printed the impeller and housing, manufactured a custom aluminum shaft, and integrated a brushless DC motor; optimized impeller geometry using fluid dynamics simulations.
- Designed analog driver for the pump (555-timer PWM + MOSFET H-bridge) to control speed/direction.

Rocket Nozzle and Ignition Control System | BU Rocket Propulsion Group

Sep 2024 - Dec 2024

- Modeled multiple fixed-geometry rocket nozzles, achieving a simulated 12% performance gain over baseline designs; ran thermal-load FEA simulations and 3D printed the top design for experimental validation.
- Engineered the test system in C++, integrating solenoid valve and electronic match with a microcontroller for synchronized fuel delivery and ignition, which enabled testing of the nozzle at target conditions.

LEADERSHIP & ACTIVITIES

Society of Hispanic Professional Engineers | Active Member and Participant

Oct 2024 - Present

USFCA and BU Chess club | Club Co-founder and Competitor (FIDE Rating: 1704)

Sep 2022 - Present

SKILLS

- Mechanical:** Lathe, Milling, 3D Printing, and Laser Cutting.
- Software:** SolidWorks, Fusion, MATLAB, Python, and embedded systems (Arduino, ESP32, STM32)
- Languages:** Fully bilingual in Spanish and English